

ConGAT: Context-Aware Graph Attention for 3D Region Discovery in Multiplexed Microscopy Images

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Abstract—The global incidence of skin cancer is rising, underscoring the growing importance of early detection. Understanding the spatial distribution of tumor and immune cell subsets in suspicious tissues is essential for researchers studying carcinogenesis progression and the effects of treatment. New multichannel microscopic imaging methods, such as Cyclic Immunofluorescence (CycIF), enable precise and reliable single-cell mapping of biomarkers in tissue biopsies. Researchers often analyze 2D microscopic images with Artificial Intelligence, but current methods rely heavily on time-consuming manual annotations and ignore 3D information. These limitations constrain biological insights and the identification of Regions of Interest (ROIs). To address these limitations, we propose ConGAT, a 3D graph attention network (GAT) that uses self-supervised refinement of positional, morphological, and intensity features to identify meaningful ROIs in 3D microscopy image volumes. It eliminates the need for labeled data, leverages depth information often obscured in 2D images, and generates 3D spatial graphs from image volumes. The model identifies ROI representations by analyzing biomarker proximity, presence, and intensities. This multi-factor analysis enables the automatic recognition of significant regions that might be overlooked by alternative methods owing to crowding, occlusion, or tissue heterogeneity. Its adaptable design enhances performance across diverse datasets and microenvironments.

Index Terms—biomarker discovery, pattern recognition, visual analytics, graph attention network (GAT), self-supervised learning, region of interest (ROI), explainability modeling.

I. INTRODUCTION

Recent advances in multiplexed microscopy imaging have become a central component of modern disease diagnosis and biomedical research, supporting detailed consideration of tissue structure and cellular organization. Multiplexed microscopy imaging techniques enable researchers to measure dozens of molecular biomarkers at cellular and subcellular resolutions [1], [2]. Compared to traditional single-marker imaging techniques such as MxIF and MIMB [3], [4], multiplexed approaches such as t-CycIF [5] and CODEX [6] provide high-dimensional spatial and cellular information and a more comprehensive view of tissue structure and organization in a single experiment [5], [6]. Additionally, these methods

provide contemporary insights into tissue-level disease processes, and the organization of the microenvironment [5], [7]. These capabilities enable experts to identify and detect regions and objects of interest that possess biologically meaningful microenvironments. These regions are important for experts to understand disease mechanisms, diagnose, and make treatment decisions [7]. However, analyzing multiplexed microscopy images remains challenging due to their high dimensionality, multi-channel nature, and limited availability of reliable annotations for supervised learning [8], [9].

First, many prior approaches rely on simple histogram-based or pairwise statistics to identify biomarker interactions, which fail to capture the joint behavior of multiple biomarkers in complex regions [10], [11]. Moreover, most meaningful interactions require co-expression and higher-order relationships within specific ranges of proximity and intensity [12]–[14]. Traditional statistical methods are limited in their ability to consider and integrate these aspects in three-dimensional multiplexed images [8], [15]–[17]. Furthermore, some leading approaches operate on two-dimensional representations of tissue sections due to the complexity and computational demands of three dimensions, limiting their ability to capture accurate three-dimensional information with depth in tissue, which most experts believe is crucial for analyzing complex microenvironments such as tumor-immune niches [13], [18], [19]. Expanding analysis from 2D to 3D introduces challenges, including higher computational demands, complex higher-dimensional relationships, and difficulties in managing the entire volumetric tissue. Additionally, recently developed approaches rely heavily on multiple datasets and manual annotations to train and validate their deep learning methods, both of which are costly and time-consuming. Moreover, locating these datasets within biomedical resources can often be challenging. In the real world, there are many tissue types and diseases, making it challenging to find comparable datasets. These dataset limitations motivate the need for self-supervised learning strategies [9], [20], [21]. Lastly, many current approaches are only loosely connected with 3D interactive visualization. This lack of tools limits the ability to

manipulate biomarkers and their intensities, restricting domain experts from interpreting results, validating detected regions, and exploring biological hypotheses effectively within a 3D workflow [11], [22]

To address these limitations, we combine complementary methods into a unified pipeline. This approach enables the joint consideration of multiple attributes while preserving interpretability and supporting expert-led exploration and validation. We introduce Context-Aware Graph Attention for 3D Region Discovery (ConGAT), a self-supervised pipeline of graph attention networks (GATs). This pipeline identifies biologically significant Regions of Interest (ROIs) in each microenvironment by considering related biomarkers, intensities, and their spatial proximity. Our method comprises three main components designed to overcome existing limitations. Beyond prior 2D histogram-based approaches, ConGAT explicitly models multi-marker interaction in 3D. The ConGAT leverages a multi-layer, multi-head graph attention network formulation to learn higher-order biomarker interactions directly from 3D tissue structure. This design identifies complex microenvironments. The volumetric image is partitioned into 3D supervoxels and represented as a spatial graph, where nodes encode category-aware multi-biomarker features and edges capture three-dimensional spatial proximity. It captures coordinated activity across multiple biomarkers. It avoids reliance on simplified intensity summaries or slice-wise analysis. Furthermore, ConGAT employs a self-supervised learning refinement strategy over Graph attention networks, eliminating the need for manually annotated datasets that are often resource-limited. Supervision comes from data by creating augmented views of the same spatial graph through node and channel masking and edge dropping. The self-supervised part uses contrastive learning to enforce invariance across these views. It also reconstructs masked features to reduce noise and outliers. Distance regularization helps maintain the geometry of 3D tissue organization. Together, these objectives produce stable node embeddings and category-specific saliency scores without labeled data. Most methods lack integrated visualization, requiring separate frameworks for experts to inspect results. ConGAT integrates its output into Vitesce [23], an interactive visualization tool that supports expert exploration and validation. ROIs are overlaid on a 3D image in Vitesce, allowing users to manipulate biomarkers by selecting channels, adjusting intensities, and navigating interactively. This coupling between model outputs and visualization improves interpretability, aiding biological hypothesis generation. Expert users facilitate easier pipeline evaluation.

Together, across synthetic and expert-guided evaluation on real melanoma CycIF data, ConGAT offers an interpretable, scalable, and annotation-free pipeline for identifying biologically significant regions in various microenvironments within complex 3D multiplexed microscopy images.

RELATED WORK

Several studies have explored methods to identify biologically significant ROIs and spatial domains within immune-

tumor microenvironments using multiplexed imaging and spatial analysis. Early research described immune-tumor interactions as spatially organized microenvironments that promote cancer progression and immune responses [24]. High-dimensional multiplexed imaging technologies, including CycIF and imaging mass cytometry, enable single-cell-resolved analysis of tumors and immune compartments [3], [5], [25], [26]. Later research characterized coordinated cellular neighborhoods as specialized regions linked to anti-tumor immune activity [13]. In melanoma, Nirmal et al. mapped tumor progression and immunoediting by analyzing spatial immune niches at single-cell resolution [14].

Most ROI discovery methods are fundamentally 2D, based on planar imaging and section-based pipelines [10], [27]. They rely on 2D neighborhoods and intensity features, limiting their ability to capture 3D tissue structures and interactions across depths [28]. As a result, complex 3D immune-tumor microenvironments may be fragmented or missed. ConGAT accounts for and develops a 3D-aware model that explicitly captures spatial context.

Graph neural networks (GNNs) effectively model biological data by encoding relationships between entities. Graph attention networks (GATs), a new approach in GNNs, use learnable attention to weight neighbour contributions, thereby enhancing node representation for capturing heterogeneous, context-dependent information interactions [?]. Researchers have applied several attention-based models in biological research, including protein function prediction, where graph convolutions integrate spatial and biochemical relationships within a radius [?], [29]. Spatial omics analysis employed a GAT-based architecture to identify spatial domains and cell niches, as demonstrated by STGATE, SpaGCN, and SPACEGM, along with their related frameworks [15], [16], [30]–[32]. Extension with multi-level or co-attention mechanisms further enhanced the modeling of intricate cell-cell interactions [33], [34]. Despite advances, most GAT-based approaches focus on 2D data or predefined graphs, rather than directly on high-dimensional 3D volumes or supporting interactive validation. Although GATs can model complex spatial interactions, their application is limited by the scarcity and high cost of annotations, which encourage the use of self-supervised learning (SSL). SSL addresses annotation scarcity by deriving supervisory signals directly from data and learning without explicit labels [9]. In biomedical imaging, SSL has been applied to histopathology and tissue analysis, allowing the model to learn semantic structure from patch-level labels [18]. SSL paradigms have also demonstrated effectiveness in learning structure-aware representations in biological domains, such as protein modeling and subcellular phenotype discovery, highlighting their ability to uncover latent structures [35], [36]. SSL is combined with graph neural networks to help retrieve ROIs in histopathology, using contrastive objectives to ensure consistency across related graph regions [21]. While prior SSL approaches focus on 2D or predefined graphs with a single objective, we integrate contrastive, masking, and distance-based learning to enforce 3D spatial coherence and improve

ROI identification in GATs.

Visual analytics play a critical role in expert-driven validation and investigation of computational findings in multiplexed microscopy images. Several visualization frameworks, including Napari, Viv, and Voreen, enable interactive visualization of microscopy images [37]–[39], but remain largely separate from computational model outputs, limiting interpretability and biological validation [40]. Frameworks such as Vitessce [23] allow researchers to overlay coordinates onto the original volumetric image. ConGAT uses Vitessce for validation and exploration, enabling users to overlay detected ROIs onto original volumetric images, thereby clearly visualizing microenvironment-specific regions within natural tissue.

III. METHODOLOGY

Multiplexed 3D CycIF volumes reveal tumor-immune microenvironments across multiple biomarker channels. Pathology experts identify relevant ROIs based on the combined effects of marker co-expression, intensity, and spatial proximity [12]–[14]. In ConGAT, we have developed a pipeline that first transforms each volumetric image into a 3D spatial graph by supervoxelizing the tissue and creating category-aware node features with radius-based edges that preserve local tissue topology. Next, we have developed an attention-based graph encoder that learns contextual embeddings by selectively weighting neighboring regions according to their relevance, enabling the modeling of higher-order interactions beyond simple pairwise statistics. We enhance the GAT encoder with a self-supervised training stage that operates on augmented graph views. This process enforces spatial coherence, resulting in improved embedding quality and more accurate saliency estimation, without relying on ROI annotations. Finally, we have designed an ROI scoring and extraction stage, in which we rank the top ROIs by their scores in interactions within their category-specific biomarker signatures for expert analysis. Fig. ?? shows how ConGAT builds the spatial graph, enhances the GAT with self-supervised learning, and detects ROIs.

A. Spatial Graph Construction

The goal of this stage is to convert high-dimensional CycIF tissue volumes into a spatial graph representation suitable for graph attention learning. Each image volume $V \in \mathbb{R}^{T \times C \times Z \times Y \times X}$ is represented as a five-dimensional tensor, where C denotes the number of biomarker channels and each voxel value $V(t, c, z, y, x)$ corresponds to the measured fluorescence intensity of channel c at spatial coordinate (z, y, x) and time point t . While time is fixed in the current analysis, the proposed framework is readily extensible to time-series imaging data.

Microenvironment categories are defined by biological parts and grouped according to biomarker expression profiles, as summarized in Table 1. These categories reduce feature dimensionality by consolidating biomarkers associated with each microenvironment while preserving biologically meaningful structure. Each category C_k represents a coherent

phenotypic microenvironment. For each node i , features are encoded using a category-aware structural representation that retains individual biomarker intensities along with their corresponding spatial locations.

$$\mathcal{C} = \{C_1, C_2, C_3, \dots, C_n\} \quad (1)$$

To reduce computational complexity while preserving 3D tissue structure, we partition the volume into contiguous 3D supervoxels, each represented by S_i , a block of $n \times n \times n$ voxels. The size n corresponds to biologically relevant interaction scales within the microenvironment. The choice of n is guided by biologically meaningful interaction scales of 3–5 μm , and given the 0.1–0.3 μm voxel resolution in CycIF imaging [12], [14], [41], we set $n = 16$ to balance biological fidelity and computational efficiency. Each supervoxel combines local voxel data into a single node, allowing scalable analysis while preserving spatial coherence. Supervoxels are nodes, and links between biomarkers are edges in the GAT.

$$\mathcal{S} = \{S_1, S_2, \dots, S_n\} \quad (2)$$

The spatial location of each node i is represented by its centroid S_i , with geometric center (x_i, y_i, z_i) , and the local intensity at each node is computed by averaging voxel-level intensities within the supervoxel independently for each biomarker channel c . For each S_i and category C_k , the node feature is defined as $F_i^{(k)}$.

$$f_i^{(k)} = \{(c, \bar{I}_{i,c}, \mathbf{X}_i) \mid c \in C_k\} \quad (3)$$

where each biomarker within the category contributes a distinct intensity value while sharing the same supervoxel centroid. If category C_k contains k biomarkers, then $f_i^{(k)}$ consists of k biomarker-specific features anchored to a common spatial coordinate. Here, $f_i^{(k)}$ summarizes the expression strength of category k at node i . The $\mathbf{F}_i$ is a complete node feature representation.

$$f_i^{(k)} = \{(c, \bar{I}_{i,c}, \mathbf{X}_i) \mid c \in C_k\} \quad (4)$$

Each supervoxel is linked to a local spatial neighborhood that captures mid-range tissue interactions. The neighborhood of node i comprises all supervoxels with centroids within a 30 μm radius, approximately twice the supervoxel length, ensuring a balance between biological relevance and contextual coverage. Edges connect nodes, and their weights encode biomarker-aware spatial relationships. The edge weight w_{ij} between distinct supervoxels i and j is defined by an inverse-distance relationship.

$$w_{ij} = \frac{1}{\|\mathbf{X}_i - \mathbf{X}_j\|_2 + 1} \quad (5)$$

This scheme enforces strong biomarker coupling within supervoxels and weakens interactions between neighboring supervoxels with distance. The graph encodes biomarker expression and spatial proximity, allowing the GAT encoder to learn context-aware tissue representations. Each supervoxel

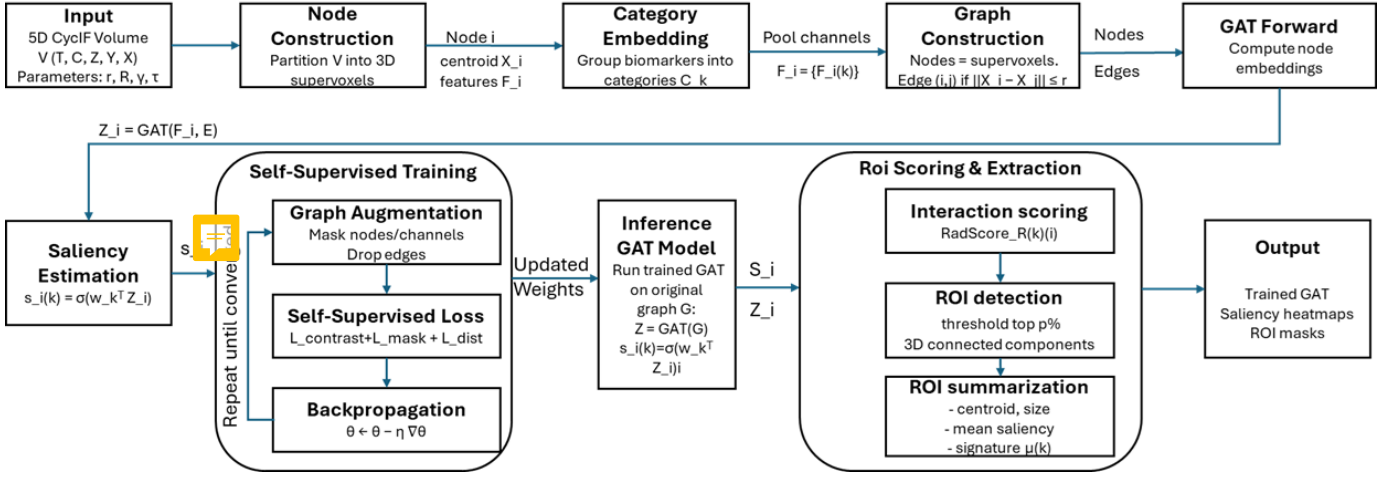


Fig. 1. Overview of the ConGAT framework illustrating graph construction, self-supervised graph attention training, and three-dimensional region-of-interest inference.

i has a composite feature vector that combines biomarker features and spatial information as input to the GAT encoder.

$$\mathbf{F}_i = [f_i^{(1)}, f_i^{(2)}, \dots, f_i^{(K)}, \{w_i\}] \quad (6)$$

where $\mathbf{F}_i$ denotes the biomarker feature of nodes and provides a unified GAT-ready encoding of local biomarker expression and spatial tissue context.

B. Graph Attention Encoding

The method uses a shared GAT to learn spatial context beyond individual supervoxels, integrating features and neighborhood data. Its attention mechanism considers biomarker patterns, categories, and spatial proximity through edge weights. Given the spatial graph $G = (V, E)$ and vectorized node representations $\mathbf{F}_i$, a GAT encoder computes a contextual embedding $\mathbf{Z}_i$ for each node.

$$\mathbf{Z}_i = \text{GAT}(\mathbf{F}_i, \mathcal{E}) \quad \mathbf{Z}_i \in \mathbb{R}^d \quad (7)$$

Here, $\mathbf{Z}_i$ denotes the learned embedding of node i , while $\mathcal{E}$ represents spatial connectivity within a radius r . The GAT computes attention coefficients over each node's neighbors to weight neighbor contributions based on biomarker intensity, distribution, number, and inverse-distance edge weights. This captures meaningful tissue interactions. To achieve interpretable, category-specific importance scores, node embeddings are mapped to saliency values using lightweight linear heads for each category k .

$$s_i^{(k)} = \sigma(\mathbf{w}_k^T \mathbf{Z}_i), \quad s_i^{(k)} \in [0, 1] \quad (8)$$

where $\mathbf{w}_k$ is a learnable parameter vector that encapsulates the interaction strength for category k . These weights reflect patterns of biomarker intensities and the composition of categories within the node embeddings. The sigmoid function $\sigma(\cdot)$ maps the scores to the range $[0, 1]$, thereby producing saliency values for attention maps over tissue using the corresponding spatial coordinates and scores.

C. Self-Supervised Graph Encoder Training

Our objective is to train the GAT encoder without manual ROI annotations by leveraging the intrinsic spatial structure and biomarker co-expression patterns. The self-supervised training stage prepares the model for downstream ROI scoring by encouraging robust, spatially coherent, and biologically meaningful node representations. A single shared GAT encoder is trained on sample subgraphs, ensuring consistent representation learning across the tissue volume. Self-supervised optimization is performed over multiple training iterations using a dedicated training split, and performance is evaluated on held-out data during inference. To generate multiple perturbed views of the same spatial graph while preserving biological identity and introducing controlled variability. The SSL model uses two augmented graphs.

$$G^{(1)} = \text{Augment}(G), \quad G^{(2)} = \text{Augment}(G) \quad (9)$$

The augmentation operator applies node-feature masking, biomarker-channel masking, and edge dropping to simulate missing or weak biomarker signals, while perturbing local connectivity. These augmentations encourage the encoder to rely on stable cues rather than noisy features. Each augmented graph is processed independently by the same GAT encoder, producing node embeddings.

$$\mathbf{Z}_i^{(1)} = \text{GAT}(\mathbf{F}_i^{(1)}, \mathbf{E}^{(1)}), \quad \mathbf{Z}_i^{(2)} = \text{GAT}(\mathbf{F}_i^{(2)}, \mathbf{E}^{(2)}) \quad (10)$$

To guide self-supervised refinement, we combine learning objectives that enforce invariance, robustness, and spatial coherence. A contrastive objective aligns embeddings across graph views to capture stable biological patterns. A masked-feature reconstruction loss promotes resilience to missing or noisy signals, and spatial distance regularization preserves tissue geometry. Together, these enable the GAT to learn meaningful and spatially consistent representations without manual annotations.

1) *Contrastive Consistency*: To ensure invariance to graph perturbations, we train the encoder to produce similar embeddings for corresponding nodes across augmented views while distinguishing them from other nodes within the batch, using a contrastive objective with temperature τ .

$$\mathcal{L}_{\text{contrast}} = -\log \left[\frac{\exp(\text{sim}(\mathbf{Z}_i^{(1)}, \mathbf{Z}_i^{(2)}) / \tau)}{\sum_{\mathbf{Z}_j \in \mathcal{B}} \exp(\text{sim}(\mathbf{Z}_i^{(1)}, \mathbf{Z}_j) / \tau)} \right] \quad (11)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and $\mathcal{B}$ denotes a batch of node embeddings. This objective encourages the same supervoxel to be embedded consistently across different perturbations, thereby promoting invariance to feature masking and edge perturbations.

2) *Masked Feature Reconstruction*: To enhance robustness to incomplete biomarker data, we introduce a masked-feature reconstruction objective. During training, a subset of node features in $\mathbf{F}_i$ is randomly masked, and the GAT encoder learns to reconstruct the original features from neighboring supervoxels. Let $\hat{\mathbf{F}}_i$ denote the reconstruction predicted from the masked graph. This loss encourages the model to leverage spatial and biological context rather than relying on individual biomarker signals.

$$\mathcal{L}_{\text{mask}} = \left\| \mathbf{F}_i - \hat{\mathbf{F}}_i \right\|_2^2 \quad (12)$$

3) *Spatial Distance Regularization*: To preserve three-dimensional tissue organization, we apply a spatial distance regularization that aligns distances in the embedding space with the physical distances between supervoxel centroids, where i and j denote neighboring supervoxels. This regularization promotes spatial coherence and prevents biologically implausible discontinuities in the learned representations.

$$\mathcal{L}_{\text{dist}} = \left\| \left\| \mathbf{Z}_i - \mathbf{Z}_j \right\|_2 - \left\| \mathbf{X}_i - \mathbf{X}_j \right\|_2 \right\|_2^2 \quad (13)$$

4) *Optimization*: The self-supervised training integrates all loss components and updates the GAT parameters θ via gradient descent with learning rate η over multiple epochs. This process enables the GAT to learn robust, spatially coherent node embeddings and category-specific saliency scores, which are subsequently used for interaction scoring, region-of-interest detection, and biomarker extraction during inference. The self-supervised objectives focus on learning node embeddings $\mathbf{Z}_i$, while the saliency scores $s_i^{(k)}$ are derived from these embeddings and are not directly optimized. This separation of representation learning from downstream analysis ensures that the encoder captures broad biological features, supporting reliable saliency estimation.

$$\mathcal{L} = \mathcal{L}_{\text{contrast}}(\tau) + \mathcal{L}_{\text{mask}} + \mathcal{L}_{\text{dist}}, \quad \theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L} \quad (14)$$

D. Scoring and Region Extraction

This aggregation highlights spatially consistent regions with strong category relevance while reducing noise. Nodes are ranked by $\text{Score}_R^{(k)}$ values, and a category-specific threshold selects top $p\%$ of nodes. Connected components among

these nodes are extracted in 3D, creating ROIs $R^{(k)}$ for category k . Each ROI yields region-level representations that capture learned interaction patterns rather than raw biomarker intensities. ROIs are defined as spatially connected supervoxels with high interaction scores, with locations from supervoxel centroids. To understand biological makeup, an interaction-weighted biomarker profile is computed.

$$\mu^{(k)} = \frac{\sum_{i \in R^{(k)}} s_i^{(k)} f_i^{(k)}}{\sum_{i \in R^{(k)}} s_i^{(k)}} \quad (15)$$

This highlights biomarkers that significantly influence category-specific interactions, ensuring ROI characterization is driven by learned spatial interactions aligned with the GAT and self-supervised objectives. Each ROI is assigned a ranking score, where $|R^{(k)}|$ denotes the number of neighboring supervoxels. The first term measures interaction consistency, while the second captures the overall interaction-weighted biomarker signal. This score prioritizes ROIs with strong interactions and high biological expressiveness, aiding the selection of relevant microenvironmental regions for downstream analysis.

$$\text{Rank}(R^{(k)}) = \frac{1}{|R^{(k)}|} \sum_{i \in R^{(k)}} s_i^{(k)} \mu_i^{(k)} \quad (16)$$

ConGAT's ROI scoring transforms node-level predictions into interpretable three-dimensional regions that reflect underlying biological interactions. The framework provides three outputs: (i) saliency heatmaps that visualize category-specific importance across tissue, (ii) 3D ROI masks identifying spatially connected biological regions, and (iii) regional biomarker signatures that support interpretation, validation, and hypothesis generation.

IV. DATASET

A. Original dataset

We use a 3D multiplexed microscopy dataset of metastatic melanoma tissue from CycIF imaging. It includes a volumetric tissue slice with niches showing organized immune and tumor biomarkers [42]. The dataset records spatial and intensity distributions of over 30 biomarkers at single-cell resolution in a sample of $194 \times 5508 \times 10908$ voxels, with an approximate voxel size of $0.1\text{--}0.3 \mu\text{m}$ [26]. It is a five-dimensional image that encodes spatial, intensity, and 3D coordinates, showing the placement, intensity, morphology, and distances of intracellular biomarkers. These volumes enable the identification and understanding of the microenvironment and treatment processes [14], [26], [43]. Biology-specific microenvironments are defined by biomarkers, enabling meaningful environmental identification based on their presence and proximity. This study investigates tissue microenvironments, including immune and tumor interactions within the image volume. We select relevant categories to search for environments that contain them, highlighting them early for expert review. Four categories, each from Table I, are chosen, with at least one

Algorithm 1 ConGAT Algorithm

- Require:** 5D C_{IF} volume $V \in \mathbb{R}^{T \times C \times Z \times Y \times X}$; graph radius r ; biological weight γ ; temperature τ
- Ensure:** 3D ROI masks, biomarker signatures $\mu^{(k)}$, attention heatmaps
- 1: **Category embedding:** Group biomarkers into categories C_k and pool channels per node: $\mathbf{F}_i = \{\mathbf{F}_i^{(k)}\}$, $k \in \mathcal{C}$.
 - 2: **Node construction:** Partition the volume V into 3D supervoxels; assign each node i a centroid $\mathbf{X}_i$ and feature vector $\mathbf{F}_i$.
 - 3: **Graph construction:** Build a spatial graph $G = (\mathcal{V}, \mathcal{E})$ by connecting nodes such that $(i, j) \in \mathcal{E}$ if $\|\mathbf{X}_i - \mathbf{X}_j\|_2 \leq r$.
 - 4: **GAT forward:** Apply a shared GAT encoder to compute node embeddings: $\mathbf{Z}_i = \text{GAT}(\mathbf{F}_i, \mathcal{E})$.
 - 5: **Saliency estimation:** Compute category-wise node saliency scores: $s_i^{(k)} = \sigma(\frac{\gamma}{\tau} \mathbf{Z}_i^T \mathbf{Z}_i)$.
 - 6: **Graph augmentation:** Generate two augmented graph views $G^{(1)}$ and $G^{(2)}$ using node/channel masking and edge dropping.
 - 7: **Self-supervised loss:** Optimize the combined objective: $\mathcal{L} = \mathcal{L}_{\text{contrast}}(\tau) + \mathcal{L}_{\text{mask}} + \mathcal{L}_{\text{dist}}$.
 - 8: **Backpropagation:** Update model parameters via gradient descent: $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$.
 - 9: **Repeat training:** Repeat Steps 6–8 over batches and epochs until convergence.
 - 10: **Inference:** Apply the trained GAT to the original graph to obtain final $\mathbf{Z}_i$ and $s_i^{(k)}$.
 - 11: **Interaction scoring:** Aggregate local saliency within radius R : $\text{RadScore}_R^{(k)}(i) = \sum_{j: \|\mathbf{X}_i - \mathbf{X}_j\|_2 \leq R} s_j^{(k)}$.
 - 12: **ROI detection:** Threshold $\text{RadScore}_R^{(k)}$ (top $p\%$) and extract 3D connected components as ROIs.
 - 13: **ROI summarization:** For each ROI $R^{(k)}$, compute centroid, size, mean saliency, and biomarker signature $\mu^{(k)}$.

present. This classification offers a structured view of biomarkers, supporting category-aware modeling and interpretation of spatial-marker patterns.

TABLE I
REPRESENTATIVE BIOMARKERS FOR TISSUE CHARACTERIZATION.

Category	Representative biomarkers
T-cell Maturation	CD3E, CD4, CD8A, CD45RA, CD45RO, CCR7, CD27, CD28, TCF7
Inflammatory Zone	IFNG, TNF α , IL6, CXCL9, CXCL10, ICAM1, VCAM1, NF- κ B
Oxidative Stress Regulation	COX-IV, Catalase, SOD1, SOD2, NRF2, GPX1, PRDX1
B-cell Infiltration	CD20, CD19, CD79A, CD79B, MS4A1, CD38, MZB1

B. Synthetic Dataset

Because annotated volumetric biomedical datasets are limited and often vary in quality and annotation consistency, we

construct synthetic 3D datasets to enable systematic, reproducible evaluation of our model. All synthetic volumes are generated at the same spatial resolution and dimensions as the real data, at the supervoxel scale ($12 \times 344 \times 681$), and contain four channels corresponding to biomarker-like signals to ensure sufficient interaction and proximity between them. To ensure consistency across all experiments, voxel intensities in each channel are independently sampled from a uniform distribution over $[0, 1]$. Across all textures and patterns, channels are constructed using band-based structures rather than narrow line patterns. All bands have a fixed width of 20 voxels, and this constraint is shared across every texture type. The spatial arrangement of bands allows channel intersections, with each synthetic volume containing 1–4 intersection regions, depending on the texture’s pattern.

Each synthetic volume belongs to one of five textures. The first texture is linear, composed of four series of linear bands with slopes of 0, 1, -1 , and vertical lines in the x - y plane, producing clear band structures with channel intersections. The second texture consists of concentric ring-like bands with similar radii, identical band widths, and different center locations, resembling Olympic rings. The third texture is formed by sinusoidal channels with three frequencies and two amplitudes, and their combination, where higher frequencies and amplitudes are integer multiples of the lowest. The fourth texture is a hybrid pattern combining three parallel linear-band channels with two curved channels oriented in the opposite direction, along with two additional cloud-like random channels overlaid on the bands. The fifth texture is an oval-based pattern, consisting of two oval tubular bands sharing the same center but with different radii, with two additional randomly distributed cloud-like channels added around the tubes; the cloud radii are comparable to the band widths, while their centers and concentrations are randomly distributed.

By deliberately controlling the spatial overlap of structures across channels, the synthetic volumes contain localized regions with varying degrees of channel intersection and intensity co-occurrence. These intersection-rich areas are explicitly treated as regions of interest, defined by the number of intersecting channels and their local intensity distributions, reflecting how domain experts identify biologically relevant regions in real volumetric imaging data. Such regions emulate meaningful biomarker co-localization and interaction patterns commonly associated with pathological tissue characteristics. Moreover, the designed textures visually resemble the volumetric appearances observed in skin lesions, moles, and similar focal abnormalities, thereby enhancing the realism and interpretability of the synthetic data. This design yields a dataset that is both structurally diverse and biologically plausible, supporting systematic evaluation of interaction-driven region discovery and localization performance while maintaining close alignment with expert-driven interpretations of real image volumes.

V. EVALUATION

We evaluate the ConGAT model using two complementary strategies. First, we generate multiple synthetic three-dimensional image volumes that vary in scale and spatial patterns while remaining consistent with the original image’s characteristics, including channel identity, signal intensity, and spatial proximity. To enable objective evaluation of these synthetic volumes, we define ground truth based on biological spatial interaction patterns. For the synthetic data, we define our ground-truth (GT) coordinates based on the spatial intersections of imaging channels, representing interaction-rich regions characterized by coordinated biomarker co-expression and spatial co-localization. These regions are extensively employed to delineate biologically significant microenvironments in multiplexed imaging [12], [13]. We compare these GT coordinates with our ConGAT results and quantify model performance using the **F1 score and Intersection-over-Union (IoU)**.

Second, we engaged nine researchers with expertise in biomedical visualization and cancer biology, and we evaluated our results based on ROIs identified in their clinical workflows. To this end, we conducted a study in which experts were asked to inspect microscopy volumes in an interactive 3D visualization framework and annotate the appropriate ROIs. The selected regions were then compared with the coordinates produced by the ConGAT model, and spatial agreement was assessed using anisotropic distance.

A. Synthetic Data Evaluation

To evaluate our model’s accuracy and robustness under diverse imaging conditions, we designed an imaging properties set that includes diverse spatial scales, textures, and multiple channels, which are significant sources of variation in real-world microscopic images and are vital for assessing model robustness under authentic conditions. Fig. 2 shows synthetic volumetric models with four channels represented as colored bands. They include linear and curved stripe patterns that mimic biological tissue features, such as filamentous, cloud-like channels and concentration areas around multiple focal points—similar to heterogeneous patterns in skin lesions. Channel intensities are normalized into $[0, 1]$, allowing intersections at different distances and intensities for different texture models. We created additional synthetic datasets by independently scaling the x -, y -, and z -axes from 1 to 2 to test scalability and geometric variability. Our primary method downsamples the original image volume via supervoxelization. Synthetic image volumes are aligned with the original image volumes to reduce computational cost and ensure method consistency. This matches the spatial resolution used after preprocessing and supervoxel aggregation, serving as the target size for all synthetic volumes. Patterns include linear, nonlinear, curved, and circular bands, as well as cloud-like structures with random centers. Using this approach, we created five textured volumes, each with seven additional scales, yielding 35 synthetic datasets.

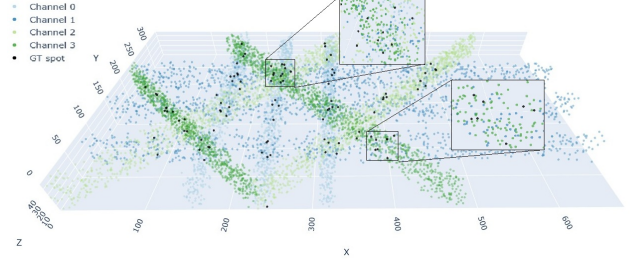


Fig. 2. 3D visualization of synthetic dataset illustrating multichannel spatial patterns and corresponding ground-truth annotations.

We evaluate the similarity between predicted and GT coordinates using voxel-based segmentation, where correspondence is quantified by the 3D anisotropic distance between the predicted and GT centers, with x and y at the exact resolution and z at half the resolution. We trained ConGAT on synthetic volumes with an 80/20 train–test split and a 10-epoch self-supervised phase on the training data. During training, contrastive objectives ensure consistency across related channels; masking objectives require reconstructing masked channels; and distance-based objectives enforce spatial coherence among neighboring voxels. After training, ConGAT makes inferences on the test set. Predicted coordinates are compared with GT coordinates using voxel-level spatial agreement; performance is measured with IoU and F1 scores computed from matches within a 3-voxel radius, with voxel differences used as the reference. The F1 score balances precision and recall, enabling objective evaluation of interaction-rich area detection.

$$\begin{aligned} \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \\ \text{F1 Score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \\ \text{Accuracy} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \end{aligned} \quad (17)$$

Here, TP is the corrected predicted coordinates, FP are predicted coordinates not in GT, FN are GT coordinates missed by the prediction, and TN are background and ignored in IoU. This complements the F1 score by emphasizing the spatial consistency of predicted interaction regions.

B. Controlled Ablation Evaluation

To directly assess whether ConGAT captures the core visual cues it is designed to model, we conduct a controlled ablation-style evaluation using simplified synthetic 3D volumes, as illustrated in Fig. 3. These volumes isolate two key parameters: channel intersection with localized intensity and channels’ spatial proximity. All samples preserve the original data’s spatial dimensions and four-channel configuration, restricted to a fixed z -plane to ensure precise ground-truth delineation and visibility to the naked eye. The linear pattern in Fig. 3(a) consists of four straight channels whose intersections define interaction-rich regions, with signal intensity locally increased

within a fixed radius around each intersection, enabling simultaneous evaluation of channel count, co-localization, and intensity sensitivity. The sinusoidal pattern in Fig. 3(b) is designed to evaluate proximity-driven interactions. Channels with varying amplitudes, periods, and small spatial shifts are arranged so they approach within a two-voxel distance, matching the graph neighborhood radius, without direct intersections.

We apply the ConGAT model, trained on synthetic image volumes, to synthetic ablation samples in inference mode and compare the predicted regions with the known ground truth. Results show that the model accurately localizes intersection-driven regions with high precision, with minor spatial offsets attributable to the three-voxel graph construction used to define meaningful interactions. In the sinusoidal sample, the model consistently identifies proximity-based interaction regions even in the absence of direct intersections, demonstrating robust sensitivity to spatial organization. Together, these controlled evaluations confirm that ConGAT effectively captures channel intersection, channel count, intensity variation, and spatial proximity as independent and complementary signals, supporting its reliability in more complex biological imaging scenarios where these cues coexist.

C. Expert Evaluation Approach

Although synthetic baselines allow for controlled evaluation, they cannot fully capture the biological complexity of real tissue microenvironments. To assess the visual and biological relevance of ConGAT in realistic conditions, we conducted an expert-guided evaluation using an interactive visualization framework on the original multiplexed melanoma image volume [44]. Nine domain experts, comprising six visualization researchers and three cancer biology researchers, examined three-dimensional tissue volumes and identified regions of interest for further investigation. Experts selected regions based on the associated number of biomarker channels, spatial proximity and intersection, and signal intensity, focusing on areas with coordinated biomarker activity and biological significance, such as immune niches and immune-tumor microenvironments. The selected coordinates served as reference points for biologically meaningful spatial locations and were compared with model output coordinates.

Spatial agreement between expert-selected regions and model predictions is quantified using an anisotropic distance measure that accounts for non-uniform voxel resolution. Smaller distances indicate greater agreement. This assessment provides a visual evaluation of ConGAT’s ability to localize interaction-rich regions.

The physical experts’ localization accuracy of predicted interaction regions is addressed by incorporating anisotropic voxel spacing during centroid localization. This approach ensures that positional errors are measured in physical spatial distances rather than in voxel indices. Anisotropy arises from differing voxel resolutions across spatial axes, necessitating

axis-specific scaling of coordinate differences.

$$d = \sqrt{((x_m - x_g)R_x)^2 + ((y_m - y_g)R_y)^2 + ((z_m - z_g)R_z)^2} \quad (18)$$

where R_x , R_y , and R_z denote the voxel resolutions along the x -, y -, and z -axes, respectively. In CycIF imaging, the in-plane X - Y resolution exceeds the through-plane Z resolution, yielding anisotropic voxels with $R_z = 2R_x = 2R_y$.

D. Experimental Validation

Experiments were conducted to evaluate the proposed ConGAT framework using 35 synthetic volumetric datasets, generated from five distinct texture types and seven spatial scales, as well as one real high-density image volume, with an 80/20 train-test split and a 100-epoch self-supervised training phase. An IoU threshold of 0.5 and a learning rate of 0.01 were used throughout. All models were implemented in Python 3.12 using GPU-accelerated PyTorch and executed on a system equipped with an Intel® Core™ Ultra 9 185H processor (16 cores, 22 threads, 2.5 GHz), 24 GB of DDR5 memory, and an NVIDIA GeForce RTX 3050 Laptop GPU with 6 GB of VRAM, running Windows 11. Each synthetic configuration required approximately 10 minutes for training and testing, while experiments on the real dataset required approximately 20 minutes per run. Inference was also conducted on the same personal computer, which served solely as a baseline for comparative purposes. Performance metrics were assessed using anisotropic distance (AD), mean error (ME), and standard deviation (Std).

VI. RESULT

The following results present a comprehensive evaluation of ConGAT, combining controlled synthetic experiments with expert-driven validation on real 3D image volumes. Due to the scarcity of reliable voxel-level ground-truth annotations, we first employ carefully designed synthetic datasets that closely match the dimensional scale and channel composition of real volumes, ensuring our pipeline is reliable and enabling its application to real datasets. These synthetic datasets enable rigorous, reproducible assessment of model accuracy and robustness across varying spatial scales and levels of textural heterogeneity and support precise quantification of spatial neighborhood agreement. To complement the synthetic evaluation, we further assess the biological relevance and interpretability of ConGAT on real multiplexed tissue volumes through expert-guided visual analysis. This analysis involves two complementary expert groups: visualization researchers, who examine regions characterized by high marker intensity and spatial concentration, and cancer biologists, who focus on regions exhibiting strong interactions among biologically meaningful biomarkers.

A. Spatial Agreement Analysis

To quantify spatial agreement, we compare the regions predicted by ConGAT with ground-truth regions derived from the spatial intersection of imaging channels, which identify

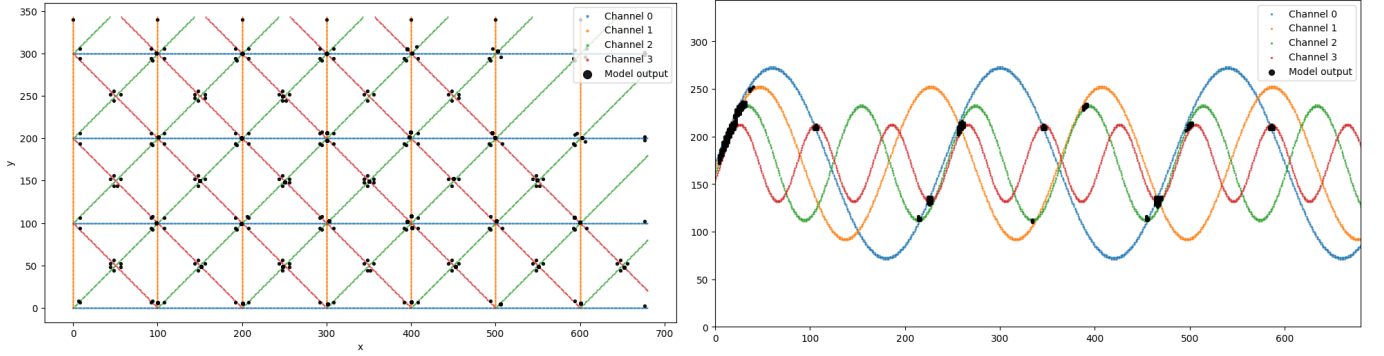


Fig. 3. Controlled synthetic samples: (a) linear channel intersections with concentrated intensity; (b) sinusoidal channels proximity-based interactions.

interaction-rich areas based on proximity, intensity, and channel count. For IoU and F1-score computation, a predicted voxel is considered a correct match if it lies within three voxels of the corresponding ground-truth voxel, thereby converting distance-based matching into a binary decision. As shown in Fig. 4, ConGAT demonstrates robust matching performance across five texture patterns and across a range of scale variations, indicating strong adaptability to spatial-scale and pattern-structure variations.

To clarify these trends, we analyze how texture periodicity and anisotropic scaling jointly affect spatial agreement. Periodic textures, such as sinusoidal patterns along the x direction, perform better when scaled along their periodic axis, as repeated intersections reinforce structural consistency and provide informative cues to the model across scales. Similarly, textures that are geometrically complete in the y direction, such as oval or Olympic-ring-like patterns, maintain continuity under y -axis scaling, yielding superior performance at that scale. In contrast, scaling non-periodic textures along a given axis primarily increases their spatial extent without adding structural information, producing spatial sparsity and degraded performance. This effect is particularly evident for x - y -axis scaling of non-periodic patterns, where expansion disrupts spatial coherence and limits the model’s ability to learn meaningful interactions.

Across all texture types, increasing scale along the z direction generally yields modest performance gains, attributable to the comparatively smaller z dimension; as depth resolution increases, the representation becomes more favorable for depth learning. These effects collectively explain the reduced F1 score and IoU observed in Fig. 4 for dense environments and regular textures at small dimensions. The most reliable performance is achieved by texture patterns that are neither completely random nor excessively sparse, but instead exhibit structured, repeatable interactions and a high ratio of active channel regions to non-channel space. Patterns such as linear arrangements and Olympic-ring-like geometries at the original scale satisfy these conditions by maintaining dense interaction regions and well-defined spatial continuity, enabling the model to capture relational features more effectively.

B. Expert Validation

We assessed the biological relevance and interpretability of regions identified by ConGAT through an expert validation study using the original melanoma tissue volume [42]. The goal was to determine whether the model-detected regions match the biologically significant microenvironments recognized by experts. Nine domain experts participated, including six computer science researchers from the University of Illinois Chicago with varied biomedical visualization experience, and three cancer biology researchers from Rush Medical University. This interdisciplinary team helped us evaluate the validity of the model’s spatial predictions by combining perceptual and biological insights.

Before the evaluation, we provided general guidance on interpreting the image volume and identifying regions using visual cues. Visualization experts examined the full tissue volume and independently selected five regions of interest based on patterns, spatial organization, and visual salience. Cancer biology experts followed a similar approach, after training on the tool’s 3D interactions, and identified five regions they deemed most biologically meaningful for validation.

To quantify agreement between expert-selected and model-predicted regions among the top five spots, we compared expert selections with ConGAT-detected regions using anisotropic spatial distance metrics. A small number of expert selections ($< 5\%$ of all selections) were identified as outliers and excluded from the analysis. For each microenvironment category, we reported the five ConGAT regions with the highest confidence, as defined by the model’s learned interaction strength. Fig. 5 shows a representative example from the Inflammatory Zone, where colored markers denote expert-selected regions and filled black circles indicate ConGAT detections. Close spatial proximity indicates strong agreement between experts and the model. Notably, some expert-selected regions appear among the top candidates with lower confidence scores yet are still detected by ConGAT, providing qualitative evidence that the model captures biologically meaningful spatial patterns consistent with prior observations in multiplexed tissue imaging.

Expert validation has been conducted across four categories, as shown in Table 1. Figure 6(a) presents inter-rater error

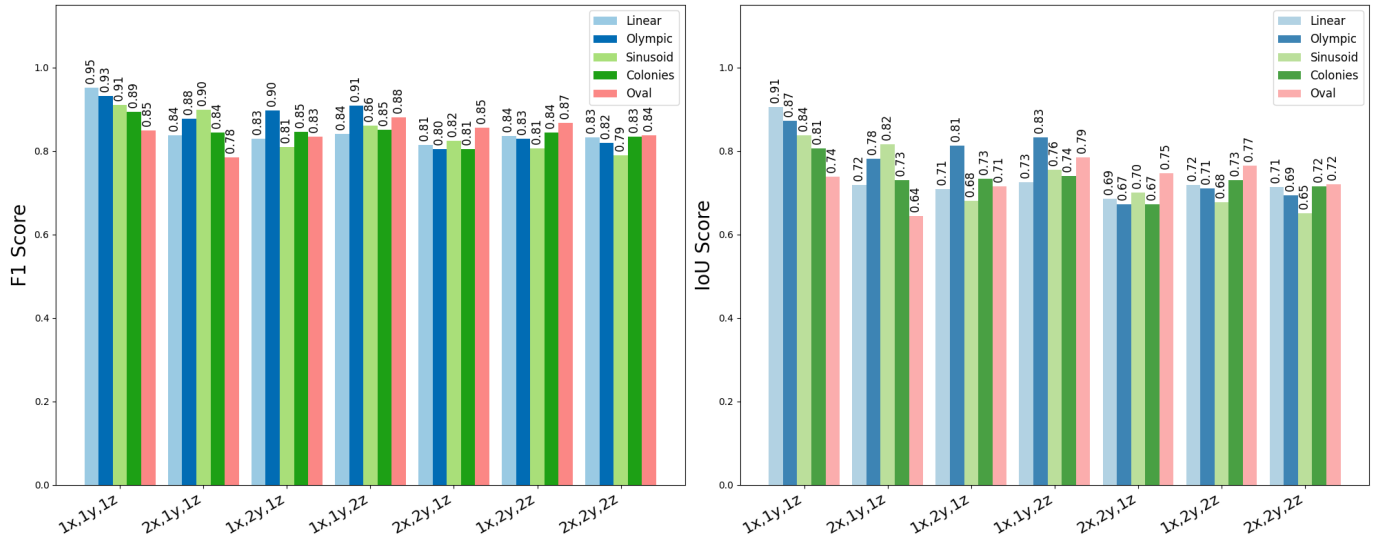


Fig. 4. Evaluation of spatial agreement: F1 score and IoU score at increasing scales for each of the five ground truth textures (linear, Olympic, sinusoid, colonies, and oval).

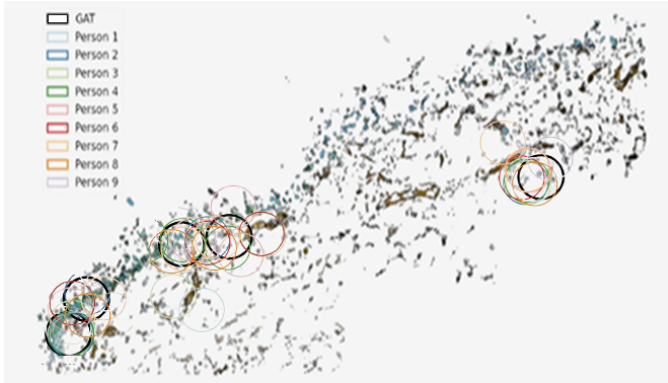


Fig. 5. 2D projection of expert regions and ConGAT detections over Inflammatory zone.

bar plots of these anisotropic distances across experts and microenvironment categories, highlighting the distribution and variability of spatial standard deviation. Lower mean distances indicate stronger spatial correspondence between expert annotations and model predictions. Complementing this analysis, Figure 6(b) presents a heatmap of median anisotropic distances, which are consistently lower than the means, indicating that most localization errors are minor and that a few larger deviations influence the mean.

Taken together, the expert-guided validation demonstrates that ConGAT robustly identifies biologically meaningful ROIs in a real three-dimensional multiplexed image volume. The model exhibits strong agreement with expert-selected regions in spatial localization, despite operating without manual annotations.

VII. DISCUSSION AND CONCLUSION

Multiplexed 3D microscopy imaging provides comprehensive insights into tissue microenvironments, enabling experts to diagnose and treat diseases more effectively. However, these images pose analytical challenges for experts in understanding all changes across the entire image due to their high dimensionality, noise, spatial and sample variability, and the scarcity of reliable 3D annotations [12], [13], [19], [25]. The ConGAT goal addresses these challenges by representing tissues as a 3D graph, where entities are modeled as nodes and links in multidimensional space. Additionally, we incorporate category-aware biomarker embeddings and employ graph attention networks to emphasize biologically relevant interactions while suppressing noise, followed by self-supervised refinement to further learn meaningful biomarker relationships. Controlled synthetic experiments show that ConGAT performs effectively across diverse spatial scales and texture conditions, achieving high neighborhood agreement and low centroid errors. Accuracy varies with structural patterns and decreases under severe anisotropic distortions, where neighborhood geometry is less reliable. Nonetheless, it maintains robust average performance across 55 synthetic volumes, demonstrating practical efficacy. Integrating ConGAT into an interactive 3D platform enhances interpretability and supports expert exploration and hypothesis generation. It also facilitates validation by revealing cluttered or overlooked regions that traditional analysis misses. Validation confirms ConGAT's ability to identify spatial regions closely matching expert-selected areas, as measured by distance metrics. While limitations remain, future work will add weak supervision, self-supervised objectives, and broader validation across tissues and imaging modalities to boost robustness and generalizability. In conclusion, ConGAT offers a self-supervised graph attention model for identifying biological regions in multiplexed microscopy. It integrates vol-

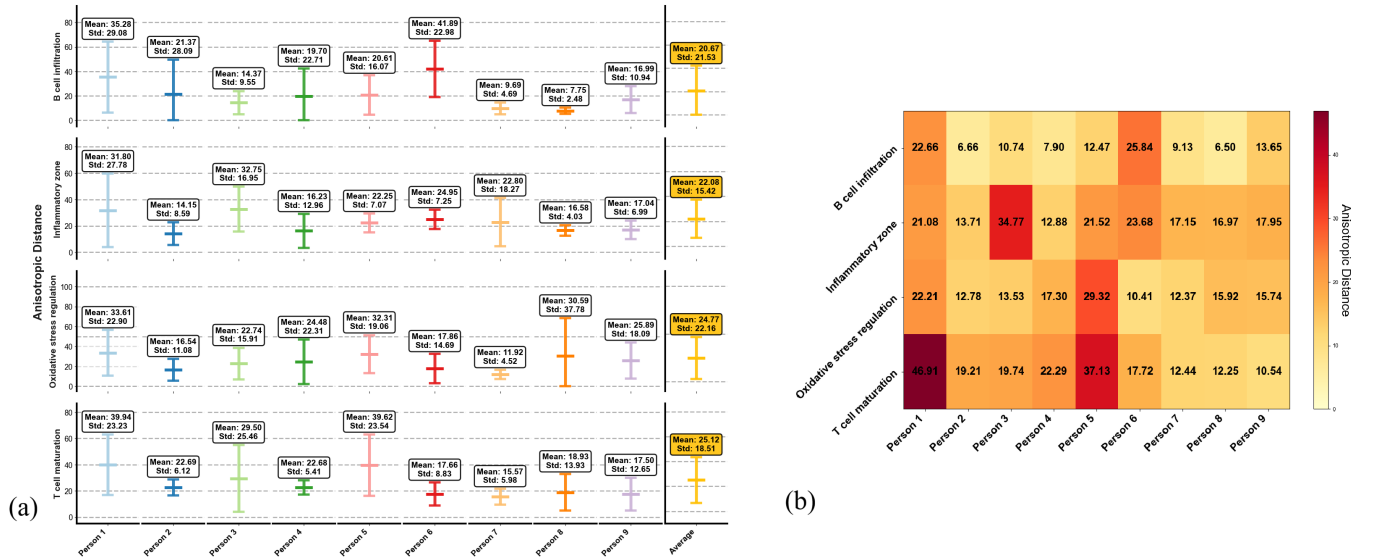


Fig. 6. Average experts' selection distance from ConGAT predictions categories:(a) inter-rater anisotropic distance bars, (b) median distance heatmap.

umetric structure, biomarker co-expression, and morphology into a single graph, overcoming annotation and 2D limitations. Synthetic tests demonstrate reliable performance across scales and levels of complexity. Expert validation confirms alignment with tumor-immune environments. Interactive 3D visualization provides a scalable, interpretable, and extensible platform for tissue analysis and pathology research.

VIII. ACKNOWLEDGEMENTS

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